

Further Differentiation

Tips, tricks and options

Key topics

Higher order differentiation

Product, chain and quotient functions

Explicit Vs. Implicit differentiation

Recap

Differentiation is the technique of altering a function into a form which describes the gradient of that function at any point

The usual notation is $\frac{dy}{dx}$ and this refers to the change in y divided by the change in x

e.g.

$$y = 4x^3 - 2x^2 + \frac{1}{x} + 2$$
$$\frac{dy}{dx} = 12x^2 - 4x - \frac{1}{x^2}$$

List of Standard derivatives

Function	Derivative
y	$\frac{dy}{dx}$
c	0
cx^n	cnx^{n-1}
$\ln(x)$	$\frac{1}{x}$
ae^{cx}	ace^{cx}
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
$a \sin(cx)$	$ac \cos(cx)$
$a \cos(cx)$	$-ac \sin(cx)$
$\tan x$	$\sec^2 x$

Double differentiation

Sometimes we want to differentiate a function more than once

Velocity differentiated is **acceleration**

To find how velocity changes with time we differentiate it

To find how acceleration changes with time, we differentiate it

$$\frac{d}{dt} \text{acceleration} = \frac{d}{dt} \left(\frac{dy}{dx} \text{velocity} \right) = \frac{d^2 y}{dx^2} \text{velocity}$$

Double Differentiation Example

e.g. $y = 4x^3 - 2x^2 + \frac{1}{x} + 2$

$$\frac{dy}{dx} = 12x^2 - 4x - \frac{1}{x^2}$$

$$\frac{d^2y}{dx^2} = 24x - 4 + \frac{1}{x^3}$$

Triple differentiation?

In fact, we can have as many as we like: $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \frac{d^4y}{dx^4}, \dots \frac{d^ny}{dx^n}$

Many functions vanish, (though some cycle between values)

e.g.

$$y = 2x^3: \frac{dy}{dx} = 6x^2, \frac{d^2y}{dx^2} = 12x, \frac{d^3y}{dx^3} = 12, \frac{d^4y}{dx^4} = 0$$

Higher order example

Find $\frac{d^2y}{dx^2}$ of: $y = \sin x + 2x^3$

$$\frac{dy}{dx} = \cos x + 6x^2$$

$$\frac{d^2y}{dx^2} = -\sin x + 12x$$

Functions of functions

$$\begin{array}{lll} \text{let} & f(t) = 4t + 5 & \text{and} & s(u) = u^3 \\ \text{then} & s(f(t)) = (4t + 5)^3 & & \end{array}$$

We **can** differentiate this function in the standard way by multiplying it out:

$$\begin{aligned} (4t + 5)^3 &= (4t + 5)(4t + 5)(4t + 5) \\ &= (4t + 5)(\mathbf{16t^2 + 40t + 25}) \\ &= 64t^3 + 160t^2 + 100t + 80t^2 + 200t + 125 \\ &= 64t^3 + 240t^2 + 300t + 125 \end{aligned}$$

Functions of functions continued

And then differentiating: $\frac{ds}{dt} = 192t^2 + 480t + 300$

$$= 2(96t^2 + 240t + 150)$$

$$= 4(48t^2 + 120t + 75)$$

$$= 12(16t^2 + 40t + 25)$$

$$= 12(4t + 5)^2$$

$$\frac{ds}{dt} (4t + 5)^3 = 12(4t + 5)^2$$

An easier way

$$(4t + 5)^3$$

We take one from the power and multiply the brackets by this:

$$3(4t + 5)^2$$

We differentiate the expression in the brackets and multiply by this also

$$12(4t + 5)^2$$

A formula for this process is:

$$s = a(bt^{n_1} + ct^{n_2} + \dots + \text{constant})^n$$

$$\frac{ds}{dt} = an(\textcolor{red}{n_1}bt^{\textcolor{red}{n_1}-1} + \textcolor{red}{n_2}ct^{\textcolor{red}{n_2}-1} + \dots + \textcolor{red}{0})(bt^{n_1} + ct^{n_2} + \dots + \text{constant})^{n-1}$$

Example 1

Let us try differentiating this expression, using the same method:

$$z = 2(3t^3 + 2t^2)^4$$

$$\frac{dz}{dt} = an(\textcolor{red}{n_1bt^{n_1-1}} + \textcolor{red}{n_2ct^{n_2-1}} + \dots + \textcolor{red}{0})(bt^{n_1} + ct^{n_2} + \dots + \textit{constant})^{n-1}$$

$$= 2 \times 4(\textcolor{red}{3 \times 3t^2} + \textcolor{red}{2 \times 2t})(3t^3 + 2t^2)^3$$

$$= 8(\textcolor{red}{9t^2} + \textcolor{red}{4t})(3t^3 + 2t^2)^3$$

The chain rule

What we have done so far is correct, but there are even easier ways to go about it!

Where $y = f(z)$ and $z = g(x)$

$$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$$

Seems a little bit easier than:

$$\frac{dz}{dt} = an(n_1bt^{n_1-1} + n_2ct^{n_2-1} + \dots + 0)(bt^{n_1} + ct^{n_2} + \dots + constant)^{n-1}$$

Doesn't it?

Chain rule example

We say let $z = 4t + 5$, $s = z^3$

We differentiate each one separately:

$$\frac{ds}{dz} = 3z^2, \quad \frac{dz}{dt} = 4$$

Now we can combine them: $\frac{ds}{dt} = 4 \times 3z^2$

And we said at the start that: $z = 4t + 5$

So: $\frac{ds}{dt} = 12(4t + 5)^2$

Second example

Try this one yourself: $z = 2(3t^3 + 2t^2)^4$, What is $\frac{dz}{dt}$?

We let $z = 2u^4$ and $u = 3t^3 + 2t^2$

We differentiate both: $\frac{dz}{du} = 8u^3$, $\frac{du}{dt} = 9t^2 + 4t$

We multiply: $\frac{dz}{dt} = \frac{dz}{du} \times \frac{du}{dt} = 8u^3 \times (9t^2 + 4t)$

We substitute u back into the equation:

$$\frac{dz}{dt} = 8(9t^2 + 4t)(3t^3 + 2t^2)^3$$

Multiplying functions

We need to be able to easily differentiate expressions, whatever their form

Consider: $y = (3x + 1)(5x - 2)$

To differentiate it, we first have to expand it:

$$y = (3x + 1)(5x - 2) = 15x^2 - x - 2$$

$$\frac{dy}{dx} = 30x - 1$$

The product rule

We'll need to differentiate some cases where we don't have the option of easily expanding the function.

$$y = u(x)v(x)$$

Where that's the case we use the product rule:

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

The product rule example

We have: $y = (3x + 1)(5x - 2)$

We let: $u = 3x + 1$ and $v = 5x - 2$

We find their derivatives: $\frac{du}{dx} = 3, \quad \frac{dv}{dx} = 5$

We apply the rule: $\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx} = (5x - 2) \times 3 + (3x + 1) \times 5$

We can write the answer as: $\frac{dy}{dx} = 3(5x - 2) + 5(3x + 1)$

Or multiply out: $\frac{dy}{dx} = 15x - 6 + 15x + 5 = 30x - 1$

Harder example

In that case, it wasn't really needed, but consider: $y = x^3 \ln(x)$

We cannot simply expand this expression, so let: $u = x^3$ and $v = \ln(x)$

Now:
$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx} = \ln(x) \times 3x^2 + x^3 \times \frac{1}{x}$$

$$= x^2 + 3x^2 \ln(x)$$

$$= x^2(1 + 3 \ln(x))$$

Your turn

Differentiate:

$$y = x^{-2}(4 + 3x^{-3})$$

We let:

$$u = x^{-2} \text{ and } v = 4 + 3x^{-3}$$

We find their derivatives:

$$\frac{du}{dx} = -2x^{-3}, \quad \frac{dv}{dx} = -9x^{-4}$$

We apply the rule:

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx} = (4 + 3x^{-3}) \times -2x^{-3} + x^{-2} \times -9x^{-4}$$

We can write the answer as:

$$\frac{dy}{dx} = -8x^{-3} - 6x^{-6} - 9x^{-6}$$

Or tidy up:

$$\frac{dy}{dx} = -15x^{-6} - 8x^{-3} = -\frac{15+8x^3}{x^6}$$

Combined example:

$$y = (x + 3)^2(4x + 1)^3$$

Let $y = uv$, where $u = (x + 3)^2$ and $v = (4x + 1)^3$

We have to do a bit of work on u and v , before we can continue

Let $u = m^2$, where $m = x + 3$ and $v = n^3$, where $n = 4x + 1$

Then: $\frac{du}{dx} = \frac{du}{dm} \times \frac{dm}{dx} = 2m \times 1$

and: $\frac{dv}{dx} = \frac{dv}{dn} \times \frac{dn}{dx} = 3n^2 \times 4$

Combined example continues

We have: $\frac{du}{dx} = 2m$ and: $\frac{dv}{dx} = 12n^2$

And using the product rule: $\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$

$$\frac{dy}{dx} = (4x + 1)^3 \times 2(x + 3) + (x + 3)^2 12(4x + 1)^2$$

This is the correct answer, but let's multiply out:

$$\frac{dy}{dx} = 340x^4 + 1728x^3 + 2628x^2 + 1010x + 114$$

You can check by multiplying out the brackets in the first expression

Quotient rule

We have looked at two functions multiplied by each other and applied the product rule

What about two functions divided by each other?

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Quotient rule: easy example

Consider $y = \frac{(x-1)}{(x+2)}$, we have two functions, one divided by the other

$$y = \frac{u}{v}, \quad u = x - 1, \quad v = x + 2$$

$$\frac{du}{dx} = 1,$$

$$\frac{dv}{dx} = 1,$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(x + 2) - (x - 1)}{(x + 2)^2}$$

$$= \frac{(x + 2) - (x - 1)}{(x + 2)^2}$$

$$= \frac{3}{(x + 2)^2}$$

Quotient rule, harder example

$$y = \frac{e^{-3t}}{t^2 + 1}$$

$$y = \frac{u}{v}, \quad u = e^{-3t}, \quad v = t^2 + 1$$
$$\frac{du}{dx} = -3e^{-3t}, \quad \frac{dv}{dx} = 2t$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} = \frac{(t^2 + 1)(-3e^{-3t}) - (e^{-3t})(2t)}{(t^2 + 1)^2} \\ &= \frac{-3t^2 e^{-3t} - 3e^{-3t} - 2te^{-3t}}{(t^2 + 1)^2} \\ &= \frac{-(3t^2 + 2t + 3)e^{-3t}}{(t^2 + 1)^2} \end{aligned}$$

Options – part 1

Consider $f(x) = \frac{1-x^2}{5x^2}$

We have a function over a function so we use the quotient rule:

$$y = \frac{u}{v}, \quad u = 1 - x^2, \quad v = 5x^2, \quad \frac{du}{dx} = -2x, \quad \frac{dv}{dx} = 10x$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} = \frac{-2x(5x^2) - 10x(1 - x^2)}{(5x^2)^2}$$

$$\frac{dy}{dx} = \frac{-10x^3 - 10x + 10x^3}{25x^4} = \frac{-10x}{25x^4} = \frac{-2}{5x^3}$$

Options – part 2

Consider $f(x) = \frac{1-x^2}{5x^2}$ as being essentially the same as: $f(x) = (1 - x^2) \frac{1}{5} x^{-2}$

We have a function **multiplied** by a function so we use the product rule:

$$y = uv \quad u = 1 - x^2, \quad v = \frac{1}{5}x^{-2}, \quad \frac{du}{dx} = -2x, \quad \frac{dv}{dx} = -\frac{2}{5}x^{-3}$$

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx} = \left(\frac{1}{5}x^{-2}\right)(-2x) + (1 - x^2)\left(-\frac{2}{5}x^{-3}\right)$$

$$\frac{dy}{dx} = -\frac{2}{5}x^{-1} - \frac{2}{5}x^{-3} + \frac{2}{5}x^{-1} = -\frac{2}{5}x^{-3} = \frac{-2}{5x^3}$$

Implicit differentiation – an extension

Explicit functions are those of the form $y = f(x)$

Differentiating explicit functions is different to differentiating implicit functions.

An implicit function may be of the form: $x^3 + 3y^2 - 4x + y = 11$.

To differentiate this you need to differentiate each part separately:

$$\frac{d}{dx} x^3 + \frac{d}{dx} 3y^2 - \frac{d}{dx} 4x + \frac{d}{dx} y = \frac{d}{dx} 11$$

Implicit example: $x^3 + 3y^2 - 4x + y = 11$

If we are differentiating with respect to x , how do we differentiate: $\frac{d}{dx} 3y^2$?

The chain rule states: $\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$, so we let $z = 3y^2$

Now: $\frac{dy}{dz} = 6y$ and if : $\frac{dz}{dx} = \frac{dz}{dy} \times \frac{dy}{dx}$

(stay with me here, we just changed the letters!)

Then: $\frac{dz}{dx} = 6y \frac{dy}{dx}$

So, when we have a function of y , we differentiate it and add $\frac{dy}{dx}$ to the end!

Therefore, $x^3 + 3y^2 - 4x + y = 11$ becomes $3x^2 + 6y \frac{dy}{dx} - 4 + \frac{dy}{dx} = 0$

We can rearrange to make $\frac{dy}{dx}$ the subject: $\frac{dy}{dx} = \frac{-3x^2+4}{6y+1}$

More help

Chapter 45, p451 in Foundation Maths, by Tony Croft

(and there are only 48 chapters – you’ve come a long way, people, I’m proud of you!)